Mechanical Ventilation – Section Summary

Troubleshooting • Asynchrony • APRV/HFOV • Weaning • NIV

Critical Care Board Review • Section 13 • MCQ-Oriented

1. Troubleshooting

- Low pressure/volume loss → **cuff rupture/leak**
- Severe obstruction + hypotension → **disconnect** briefly
- Initial settings from **ideal body weight + minute ventilation**
- Flow not reaching zero / ineffective efforts (auto-PEEP) → **↑ PEEP** (COPD) or ↑ sedation
- Scooped pressure curve (flow starvation) → **↑ flow**
- PRVC premature termination asynchrony

2. Physiology

- COPD: **high compliance, high resistance**
- Spontaneous effort ↑ transpulmonary pressure (e.g. 10 → 25)
- HFOV gas transport: pendelluft, coaxial flow, Taylor dispersion; CO_2 ↓ by ↑ amplitude/cycle volume
- Pressure SIMV → greatest breath variation

3. APRV

- ↑ CO_2 → **↑ T-low** (or ↑ P-high)
- Good oxygenation, FiO_2 40% → wean: **↓ P-high, ↑ T-high** (drop and stretch)

4. Weaning & NIV

- SBT: CPAP 5 → calculate **RSBI**
- COPD after 3 days → extubate to **NIPPV**
- Post-extubation stridor risk → **cuff-leak test**
- Strongest NIV indication: **AECOPD pH < 7.35**
- VAP: chlorhexidine oral care; prolonged weaning → trach, most eventually wean

5. High-Yield MCQ Pearls

- ☑ COPD: high compliance + high resistance
- ☑ APRV wean: drop and stretch

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